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An Interpretable and Accurate Ensemble Machine Learning Approach for Dementia Prediction

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***Abstract*—**Dementia is a significant global health issue that necessitates accurate and interpretable predictive models to aid in early detection and intervention. The work suggests a strong machine learning framework for dementia prediction that makes use of SHAP (Shapley additive explanations) values to improve model interpretability. The work uses extensive preprocessing methods to ensure data consistency and quality such as handling missing values, normalization, encoding, scaling and data balancing using SMOTE. To find the most pertinent predictors, a filter mechanism is used for feature selection. Ten machine learning models were used in this work. A voting ensemble combining logistic regression (LR) and extra tree classifier outperformed other machine learning models achieving the maximum accuracy of 98.67% using the OASIS dataset. Our system addresses the lack of interpretability in existing models and gives better performance in terms of accuracy, precision and recall than previous research. SHAP values ensured transparency and dependability in model choices by identifying important predictive features including MMSE, clinical dementia rating (CDR) and MRI related metrics like MR Delay. These findings show how our SHAP integrated machine learning system can produce precise, understandable and useful predictions, opening the door to better dementia diagnosis and individualized treatment.

***Keywords*—** Dementia prediction, normalization, machine learning models, SMOTE, SHAP, voting.

I. INTRODUCTION

Millions of people worldwide suffer from dementia, a neurological disease marked by cognitive loss and its prevalence is predicted to increase significantly as the population ages [1]. Early diagnosis is essential to enabling prompt interventions and improving quality of life since it has a substantial influence on patients, caregivers and healthcare systems [2]. However, the need for creative, precise and scalable alternatives is highlighted by the fact that conventional diagnostic techniques are subject to variability and mostly rely on clinical experience [3]. Machine learning (ML) has become a popular technology for evaluating high-dimensional, complicated datasets in recent years. ML algorithms can help with dementia subtype classification, disease progression prediction and key predictive feature identification by recognizing patterns in clinical, demographic and imaging data [4]. Despite this promise, issues including overfitting, interpretability and the requirement for trustworthy feature selection make it difficult to create strong machine learning models for dementia detection [5].

The fundamental processes of many types of dementia are still not fully understood, despite tremendous advancements in medical and neurological research [6]. Determining the underlying causes of dementia is difficult due to the complexity of the brain and the multifaceted nature of the condition which includes lifestyle, environmental, and hereditary variables [7]. The limitations of current therapy choices are highlighted by the fact that current treatments mostly concentrate on managing symptoms, ensemble approaches and interpretability techniques. Our findings show ML's potential to improve dementia care, paving the path for more targeted and efficient interventions. The following highlights the contributions that our work has made:

- To ensure data quality and robustness, provide a thorough preprocessing pipeline that addresses missing values, normalization, encoding, feature scaling, and class imbalance using SMOTE.
- The most pertinent characteristics are found using a filter-based feature selection technique, which enhances the generalization and performance of the model.
- Present a unique ensemble architecture that achieves the greatest accuracy for dementia prediction on the OASIS dataset by combining Extra Trees Classifier (ETC) and Logistic Regression (LR) through a voting mechanism.
- Find clinically significant predictors by integrating SHAP (Shapley Additive Explanations) values to guarantee interpretability and transparency.

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II. Related Work

Sara Dhakal et al. [9] used datasets OASIS machine learning (ML) techniques have demonstrated potential in the prediction of dementia. Algorithms using and without LASSO selected features including SVM, RF and gradient boost, have been investigated. SVM efficacy was demonstrated by its greatest accuracy of 96.77%. These techniques show promise for an accurate and timely diagnosis. Aram So et al. [10] using MMSE-KC and CERAD-K data, a two-layer machine learning model was created to distinguish between dementia, mild cognitive impairment (MCI) and normal. When algorithms such as SVM, random forest and MLP were evaluated, the model's F-measure was highest with MLP (0.97 for normal) and SVM (0.739 for MCI and dementia). This method cuts expenses and time while simplifying diagnostics.

Huan Wang et al. [11] tested using data from 1,100 community members and 654 nursing home residents in China. The best results were obtained by SVM with a polynomial kernel (AUROC up to 0.9435). LASSO found 17 essential characteristics that make the tool dependable, affordable and clinically applicable. Joel Nyholm et al. [12] used five machine learning algorithms and data from Swedish older individuals (60+) to investigate dementia prediction. The most accurate method was gradient boosting with a ROC AUC of 0.974 and an accuracy of 92.9%. Daytime sleep, sex, education, age, nightly awakenings and snoring were important predictors, highlighting the importance of individual and sleep related characteristics. Gopi Battineni et al. [13] used logistic regression (LR), k-nearest neighbor (KNN) and support vector machines (SVM) to create a machine learning (ML) model for dementia classification. To extract features, principal component analysis (PCA) was used. The model's accuracy, precision, recall and AUC were used to assess its performance. KNN came in second with an accuracy of 0.976 and an AUC of 0.966 while LR had the greatest accuracy (0.983) and AUC (0.997). The best classifiers for dementia prediction were found to be LR and KNN.

Fabian H. Reith et al. [14] used a GBDT model to predict future amyloid deposition (SUVR) by combining clinical and genetic variables with PET imaging features (extracted by ResNet) using ADNI data. The approach improved the ability to identify people with notable amyloid alterations, outperforming both linear regression and GBDT without imaging features. Chris A. Hane et al. [15] used deidentified clinical notes from hospital systems to improve ADRD risk prediction. With the biggest effect seen in years 3-6, including notes raised PPV from 45.07% to 68.32% at disease onset and AUC from 0.85 to 0.94.

Dea Gogishvili et al. [16] employed machine learning to predict cognitive impairment in dementia patients via CSF biomarkers. Random forest classifiers had an AUC of 0.82 and identified TNFRSF4 and TGFβ-1 as relevant signals. Although individual markers exhibited minor effects, the framework provides insights into customized dementia prognosis. S.K. Aruna et al. [17] suggests a paradigm for classifying dementia from MRI images by integrating ICA, GLCM and Gabor filters for feature selection and extraction. The efficacy of the framework was demonstrated by the high accuracy of SVM classifiers on the OASIS dataset.

III. Methodology

In **Fig. 1,** the flowchart presents a methodical approach for machine learning based dementia prediction. The dataset is first preprocessed using procedures including resolving missing values, normalization, scaling and encoding. The synthetic minority oversampling technique (SMOTE) is used to create a balanced dataset in order to remedy class imbalance. A filter-based feature selection process is used to find the most important features, improving model performance and lowering noise. The dataset is then divided into 70% training and 30% testing sets. To predict dementia outcomes, machine learning models like random forest (RF), gradient boosting (GBC) and logistic regression (LR) are developed. By combining the advantages of several models, ensemble approaches are used to increase accuracy and resilience. Metrics like accuracy, precision, recall, F1-score and ROC are used in performance analysis to assess the efficacy of the model. Lastly, to provide transparency and clinical reliability, SHAP (Shapley additive explanations) values are used to quantify the contribution of each attribute to the predictions, hence providing interpretability.

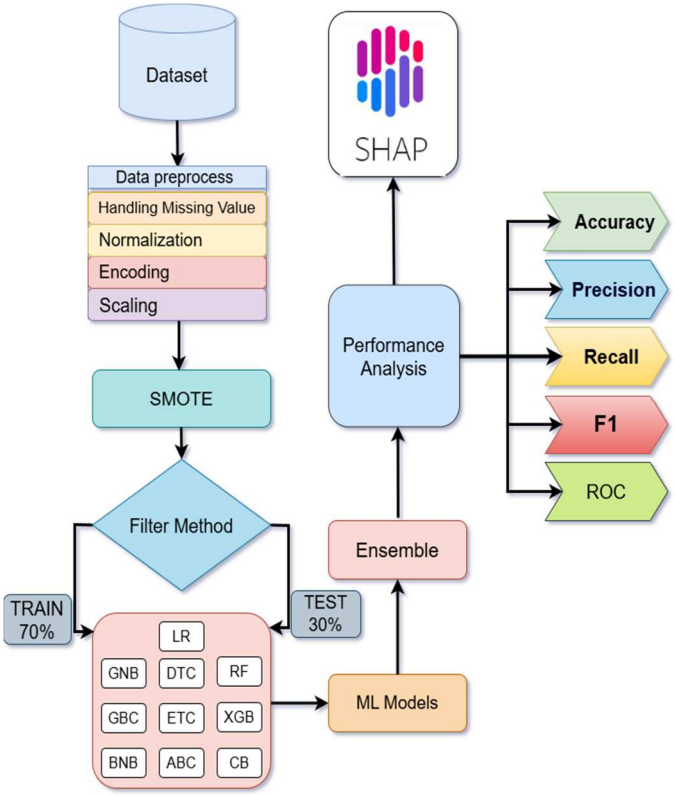


Fig. 1. Workflow of the proposed method.

A. Dataset

The dataset used in this study is a version of the open access series OASIS dataset, publicly available on Kaggle [18]. This dataset consists of 373 T1-weighted MRI images taken from 150 participants, ages 60 to 96, spanning several visits spaced at least a year apart. Of these, 64 had dementia throughout, including 51 with Alzheimer's, 72 were consistently non-demented and another 14 participants were diagnosed as nondemented at the time of their initial visit and were then characterized as demented at a subsequent visit. The dataset facilitates the development of predictive algorithms and better understanding of the evolution of dementia.

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B. Data processing

In this study, preprocessing includes converting categorical variables into numerical representations for machine learning algorithm compatibility, addressing missing values filled with the median value and normalizing features to a standard scale for uniformity. In order to correct class imbalances and create a solid and ready dataset for feature selection, training and testing, data balancing is also carried out using SMOTE.

C. Feature Selection

The most significant predictors for the desired outcome are chosen using a filter approach that assesses features according to statistical metrics like correlation, variance or significance scores. By removing unnecessary or redundant features, this phase lowers the chance of overfitting and improves the machine learning models interpretability and effectiveness. In this studies, ten features were selected using the SelectKBest filter method [19]. Select the top 10 features according to the mutual information and extra tree score in **TABLE I**.

TABLE I. FEATURE SELECTION BY MUTUAL-INFORMATION AND EXTRA TREE SCORE.

Feature	Mutual Info	Extra Trees
CDR	0.524264	0.457327
MMSE	0.293115	0.198955
nWBV	0.083074	0.045096
ASF	0.051262	0.043775
MR Delay	0.042575	0.037112
Hand	0.041087	0.000000
eTIV	0.038444	0.042622
SES	0.025674	0.031492
M/F	0.025362	0.045207
EDUC	0.022761	0.042358

D. Data Split

The dataset is split into two sections at the data split step: 30% is the testing data and 70% is the training data. While the testing data assesses the model's performance on unknown data, the training data is used to construct and refine machine learning models.

E. Proposed Hybrid Ensemble Model (LR+ETC voting)

This study presents a hybrid ensemble model that combines extra trees classifier (ETC) and logistic regression (LR) utilizing a voting method to improve interpretability and accuracy. While ETC uses randomized decision limits to capture nonlinear feature interactions, LR captures linear correlations between clinical and imaging variables. Every model in the suggested ensemble generates a class probability $P_i(y=1|x)$on its own. The probability of the final output is calculated as follows equations (1) and (2):

$$P_{final}(y=1|x) = \frac{1}{n}\sum_{i=1}^{n} p_i(y=1|x) \quad (1)$$

Where n=2 (LR and ETC).
The final prediction class is assigned as:

$$y = \begin{cases} 1, P_{final}(y=1|x) \geq 0.5 \\ 0, \quad otherwise \end{cases} \quad (2)$$

IV. RESULT AND DISCUSSION

Metrics including accuracy, precision, recall, F1-score and AUC are used to evaluate the model's performance, providing a thorough assessment of prediction efficacy and class discrimination [22]. High-performance CPUs or GPUs are used by the framework to handle massive datasets and train intricate machine learning models. we determined the performance parameter by using the following equations: (3), (4), (5), (6), and (7).

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (3)$$

$$Precision = \frac{TP}{TP+FP} \quad (4)$$

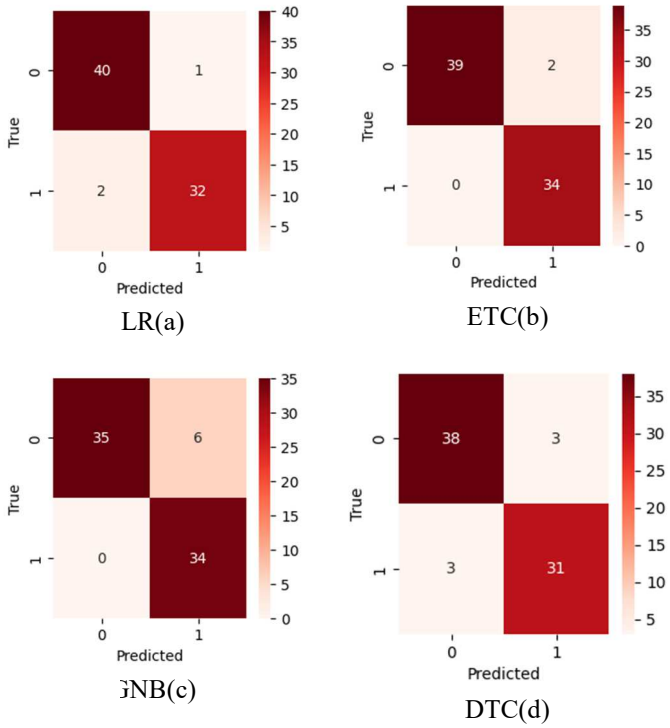
$$Recall = \frac{TP}{TP+FN} \quad (5)$$

$$F_1 - Score = 2 * \left(\frac{Recall * Precision}{Recall + Precision}\right) \quad (6)$$

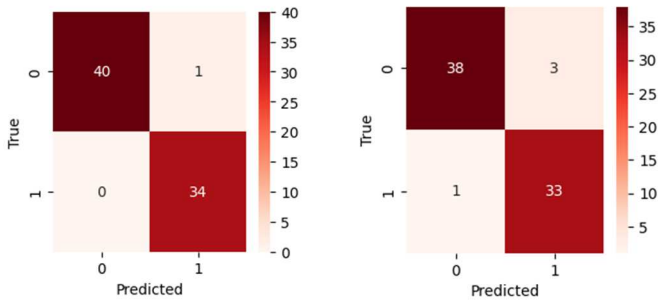
$$AUC = \sum_{i=1}^{n-1} (FPR_{i+1} - FPR_i) * \frac{(TPR_{i+1} - TPR_i)}{2} \quad (7)$$

A. Confusion Metrics

In **Fig. 2** LR+ETC(e) outperforms the other models with perfect recall, great precision and only one false positive, making it the most balanced and trustworthy. ETC(b) and LR(a) both perform well, with ETC(b) reaching perfect recall but somewhat more false positives and LR(a) demonstrating great overall balance and little misclassifications. Compared to the best models, DTC(d) has more false negatives but produces respectable results. Even while GNB(c) achieves perfect recall, its precision is greatly reduced by the highest number of false positives. Due to its unbalanced predictions, GNB(c) is the weakest model overall, while LR+ETC(e) is the most successful.



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LR+ETC(e) ETC+CB(f)

Fig. 2. Confusion metrices for ML models and Ensembles.

B. Roc Curve

In **Fig. 3** the ROC curve for the extra trees classifier (ETC) shows exceptional performance with an AUC score of 0.998, suggesting nearly flawless discrimination between positive and negative categories. The curve remains close to the upper left corner, indicating a high true positive rate and a very low false positive rate throughout thresholds. This demonstrates the ETC model's strength and reliability in properly predictions outcomes while minimizing errors.

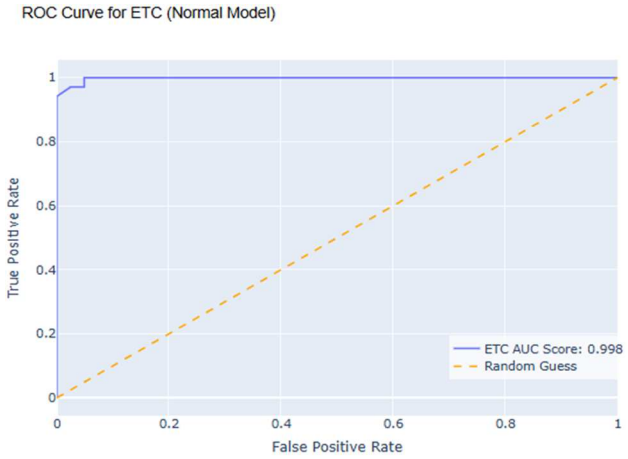


Fig. 3. Roc curve of the ETC model.

In **Fig. 4** the ROC curve for the voting classifier (LR_ETC) shows remarkable performance with an AUC score of 0.997, showing that it can accurately identify between classes. The curve closely follows the top-left corner indicating a high true positive rate and a very low false positive rate at different levels. This illustrates the LR_ETC voting model's great predictive power and reliability.

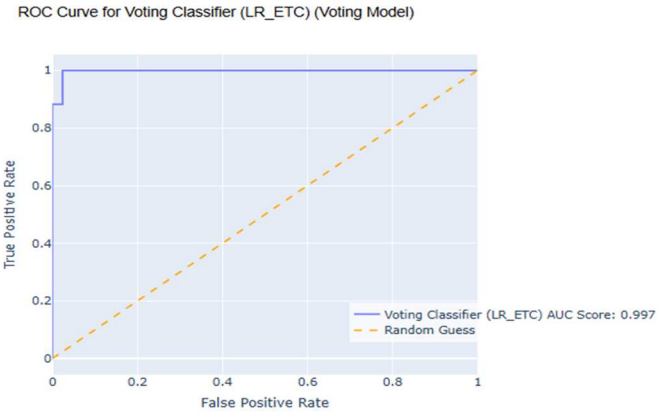


Fig. 4. Roc curve of the LR+ETC voting model

C. Figures and Tables

In **TABLE II** the most efficient model is the extra tree classifier (ETC) which has perfect recall and the highest accuracy (97.33%), F1 score (97.14%), ROC score (97.56%) and AUC (99.82%). Both GBC and RF exhibit strong performance with AUC ratings exceeding 99%. High precision and recall are balanced by LR whereas perfect recall, but poorer precision is attained by GNB.

TABLE II. MACHINE LEARNING MODEL PARAMETERS SCORE WITH FEATURES SELECTION

Model	Accuracy	Precision	Recall	F1	ROC	AUC
LR	96.00	96.97	94.12	95.52	95.84	99.07
GNB	92.00	85.00	100.00	91.89	92.68	98.71
DTC	92.00	91.18	91.18	91.18	91.93	91.93
RF	96.00	91.89	100.00	95.77	96.34	99.64
GBC	94.67	91.67	97.06	94.29	94.87	99.71
ETC	97.33	94.44	100.00	97.14	97.56	99.82
XGB	94.67	91.67	97.06	94.29	94.87	99.57
BNB	94.67	89.47	100.00	94.44	95.12	96.20
ABC	93.33	93.94	91.18	92.54	93.15	99.10
CB	93.33	91.43	94.12	92.75	93.40	99.35

In **TABLE III,** the performance of voting classifiers that combine different models with feature selection is displayed in the table. The top model is LR_ETC which has perfect recall (100.00%) and the highest accuracy (98.67%), F1 score (98.55%), ROC score (98.78%) and AUC (99.71%). In addition, ETC_RF produces excellent results with flawless recall and a high AUC (99.68%) whilst LR_RF and LR_GBC provide balanced performance with accuracy (96.00%) and AUC scores above 99%. Although their accuracy (94.67%) and AUC ratings are marginally lower, ETC_GBC and ETC_CB have strong recall (97.06%). In all metrics, LR_ETC is in the lead, with ETC_RF coming in second.

TABLE III. APPLY VOTING ON VARIOUS MODELS WITH FEATURE SELECTION

Model	Accuracy	Precision	Recall	F1	ROC	AUC
LR_ RF	96.00	94.29	97.06	95.65	96.09	99.35
LR_ ETC	98.67	97.14	100.0	98.55	98.78	99.71
LR_ GBC	96.00	94.29	97.06	95.65	96.09	99.57
ETC_ RF	96.00	91.89	100.0	95.77	96.34	99.68
ETC_ GBC	94.67	91.67	97.06	94.29	94.87	99.78
ETC_ CB	94.67	91.67	97.06	94.29	94.87	99.43

In **Fig. 5** ETC outperforms RF (96.00%) and LR (96.00%) in terms of accuracy whereas models such as GNB (92.00%) and DTC (92.00%) fall short. **Fig. 6** as the best performing voting classifier with the highest accuracy (98.67%), LR_ETC outperforms other combinations such as LR_RF and ETC_RF (both 96.00%). ETC_GBC and ETC_CB maintain their competitiveness while achieving somewhat lower accuracy (94.67%). Combining the advantages of logistic regression and extra trees, LR_ETC is the model that performs the best overall.

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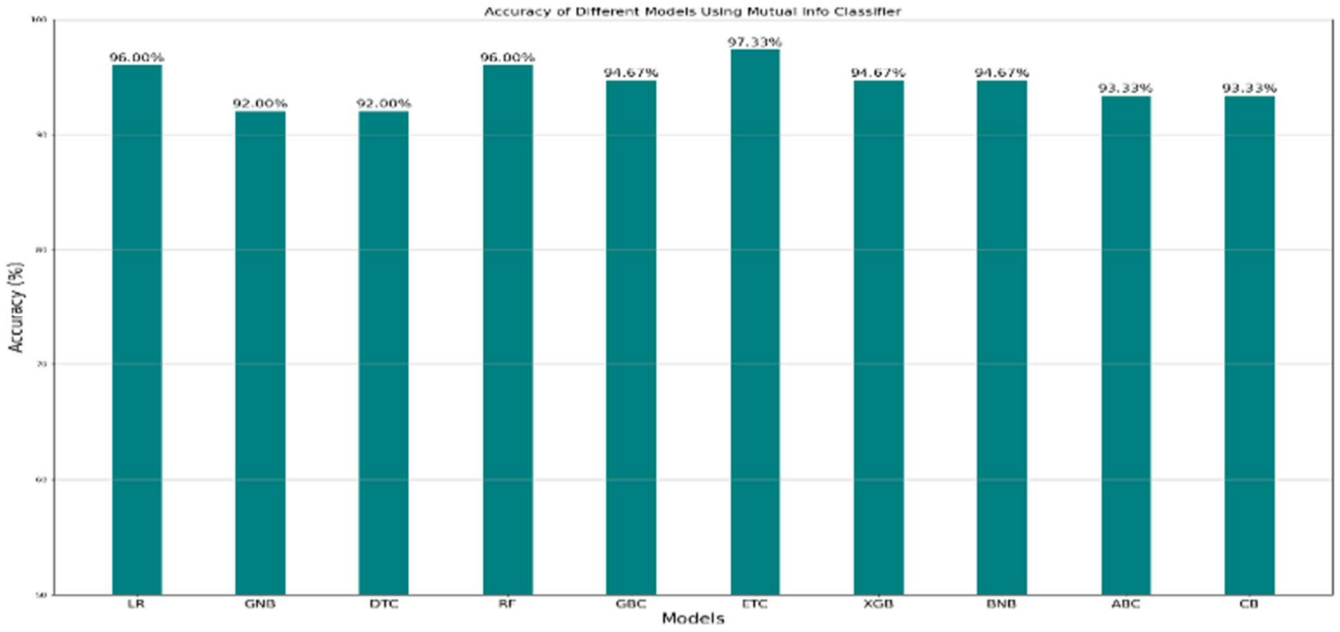


Fig. 5. Accuracy scores of ten machine learning models.

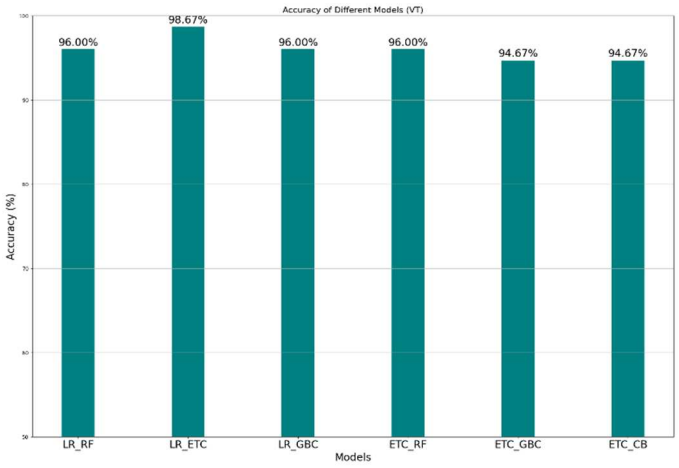


Fig. 6. Accuracy scores of the top six voting classifiers.

In **TABLE IV**, On the same dataset, the suggested LR+ETC model outperforms Gopi Battineni et al. [13] (97.60%) and Sara Dhakal et al. [9] (96.77%), achieving the maximum accuracy (98.67%) of all the works. Additionally, it performs noticeably better than others such as Huan Wang et al. [11] (88.64%) and Aram So et al. [10] (73.90%). These outcomes unequivocally show that the suggested model is the most dependable and successful, proving its superiority through feature selection and model integration.

TABLE IV. PERFORMANCE ANALYSIS WITH OTHER WORKS

Reference	Dataset	Model	Accuracy
Sara Dhakal et al.[9]	373 patient's data collected from OASIS	SVM	96.77%
Aram So et al.[10]	Data collected from MMSE-KC and CERED-k	SVM	73.9%
Huan Wang et al. [11]	Dataset collected from Hangzhou	SVM	88.64%
Gopi Battineni et al. [13]	150 Patients data collected from OASIS	KNN	97.6%
Proposed work	**373 patient's data collected from OASIS**	**LR+ETC**	**98.67%**

D. *Interpretability*

In **Fig. 7** the SHAP study shows that the two most important features that have a major influence on the model's predictions are the clinical dementia rating (CDR) and the mini mental state examination (MMSE). M/F, MR Delay and SES (socioeconomic status) are among the features that contribute moderately whereas Hand and nWBV (normalized whole brain volume) have less effect. Higher SHAP values indicate stronger feature importance and the color gradient (blue for low, red for high) shows how feature values impact predictions. In general, MMSE and CDR are the main factors influencing the model's performance.

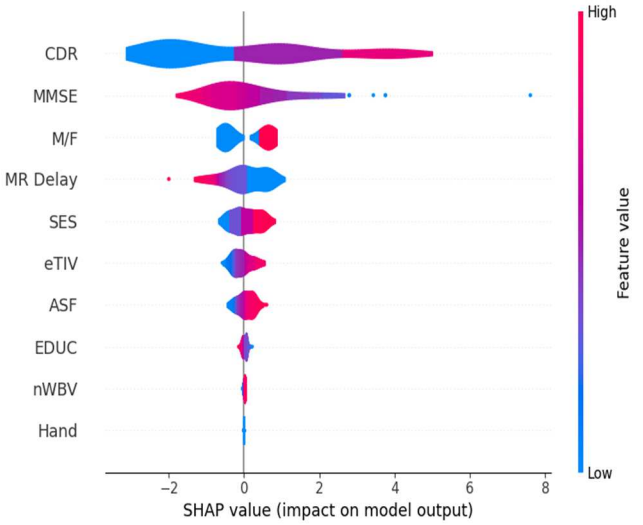


Fig. 7. SHAP analysis of features importance in model output.

In **Fig. 8** the SHAP decision plot shows CDR as the most significant feature with a large negative effect (-1.98), followed by MR Delay (-0.78) and MMSE (-0.73). M/F (+0.64) and eTIV (+0.52) make positive contributions but other aspects such as ASF, EDUC and SES have less impact. nWBV and Hand make minor contributions. The prediction is mostly driven by CDR, MR Delay and MMSE.

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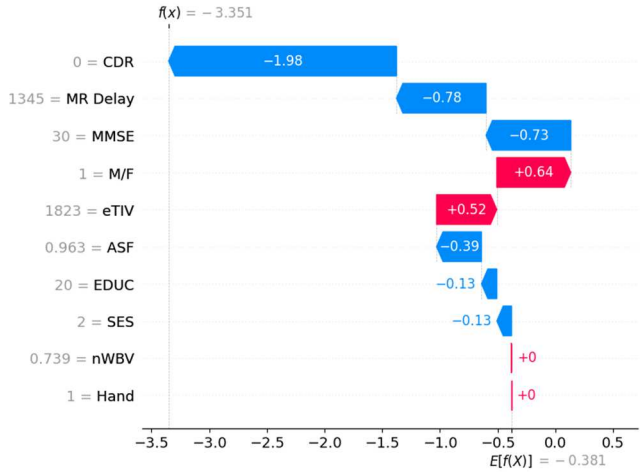


Fig. 8. SHAP decision plot key features contributes to model prediction.

In **Fig. 9** the mean SHAP values show CDR (clinical dementia rating) as the most influential characteristic with the greatest influence (+1.91), followed by MMSE (mini-mental state examination) at +0.93. M/F (+0.57), MR Delay (+0.45) and SES (+0.31) have moderate contributions, while eTIV, ASF and EDUC have a lower influence. nWBV and Hand have low influence, implying a minor contribution to the model's predictions. Overall, CDR and MMSE are the primary drivers of the model's decisions.

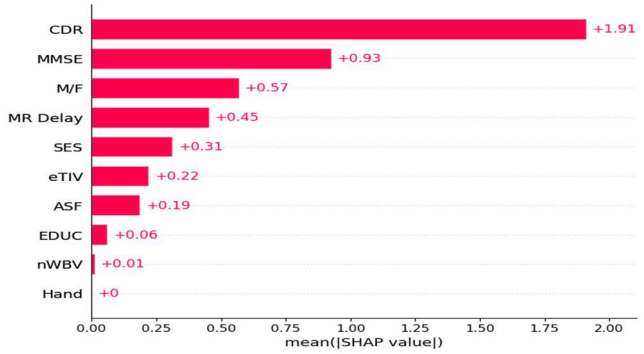


Fig. 9. Average feature relevance for SHAP-based dementia prediction.

V. Conclusion

This work shows how well machine learning models in particular, LR+ETC predict dementia based on imaging and clinical data. We outperformed other state of the art methods with our suggested framework, which had the maximum accuracy of 98.67%. The model's predictions were greatly influenced by factors such as the clinical dementia rating (CDR), MMSE and MRI related variables (MR Delay) which were highlighted by the use of SHAP values. Additionally, ensemble approaches demonstrated strong performance, confirming the dependability of integrating several classifiers. These results highlight the promise of machine learning as a scalable and precise tool for clinical applications in dementia early detection.

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